

REVIEW

Social costs of illegal drugs, alcohol and tobacco in the European Union: A systematic review

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Abstract

Issues. Drug use accounts for one of the main disease groups in Europe, with relevant consequences to society. There is an increasing need to evaluate the economic consequences of drug use in order to develop appropriate policies. Here, we review the social costs of illegal drugs, alcohol and tobacco in the European Union. **Approach.** A systematic search of relevant databases was conducted. Grey literature and previous systematic reviews were also searched. Studies reporting on social costs of illegal drugs, alcohol and tobacco were included. Methodology, cost components as well as costs were assessed from individual studies. To compare across studies, final costs were transformed to 2014 Euros. **Key Findings.** Forty-five studies reported in 43 papers met the inclusion criteria (11 for illegal drugs, 26 for alcohol and 8 for tobacco). While there was a constant inclusion of direct costs related to treatment of substance use and comorbidities, there was a high variability for the rest of cost components. Total costs showed also a great variability. Price per capita for the year 2014 ranged from €0.38 to €78 for illegal drugs, from €26 to €1500 for alcohol and from €10.55 to €391 for tobacco. **Implications and Conclusions.** Drug use imposes a heavy economic burden to Europe. However, given the high existing heterogeneity in methodologies, and in order to better assess the burden and thus to develop adequate policies, standardised methodological guidance is needed. [Barrio P, Reynolds J, García-Altés A, Gual A, Anderson P. Social costs of illegal drugs, alcohol and tobacco in the European Union: A systematic review. *Drug Alcohol Rev* 2017;00:000-000]

Key words: social cost, addiction, illegal drug, tobacco, alcohol.

Introduction

Illegal drug, alcohol and tobacco use are major contributors to the global burden of disease worldwide [1,2]. Their use does not only affect individuals alone but also societies as a whole, with consequences ranging from impaired health and premature death to social and economic decline [3].

Specifically in the European Union (EU), the magnitude of the problem is also considerable. Recent estimates suggest that more than 4% of the population is affected by alcohol or drug dependence [4]. The most recent European Drug Report issued by the European

Monitoring Centre for Drugs and Drug Addiction [5] indicated that at least one-quarter of the EU population has used illegal drugs and, according to the World Health Organization, tobacco smoking kills more Europeans than any other preventable risk factor [6]. These high figures justify the increasing efforts that have been made in recent years to accurately estimate the costs that illegal drug, alcohol and tobacco use impose on society (so-called social costs). Social costs can be understood as the sum of private plus external costs (i.e. costs that are external to the individual making the consumption, such as the costs that smoking causes to non-smokers). External costs or externalities represent almost all of the social

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Received 2 June 2016; accepted for publication 8 September 2016.

costs when assessed in substance use studies. One of the important characteristics of externalities is that they are usually borne by others than the user, hence the term ‘social’ cost. While the distinction between private and external costs is not always straightforward, under value theory premises, what users spend on drugs is supposed to be offset by the effects they obtain; hence, the cost is usually considered as zero.

Reviews on the social costs of illegal drugs, alcohol and tobacco, both globally and in the EU, suggest that these costs are high, consuming an important share of nations’ gross domestic product (GDP) [7–9].

Although the concept seems clear enough at first glance, undertaking social cost estimates has proven difficult because of methodological issues inherent to the complex phenomenon of substance use and its social consequences. Variables and causal relationships might be multiple, deferred in time and sometimes not clearly identifiable, measurable or retrievable. In many instances, for example, it might not even be straightforward whether a substance should be labelled as legal or illegal (such as cannabis). Some substances might even be used both in a legal and illegal manner (diverted prescription drugs for example, or even alcohol if consumed by underage people, according to a country’s laws). Complicating things further, there is the fast evolving situation of novel psychoactive substances or ‘legal highs’. Not only are definitions and classifications intrinsically difficult; causal connections (i.e. the extent to which a consequence or a cost can be attributed, and not only associated, to the use of a substance) present several difficulties [10,11]. This complexity, on top of considerable methodological heterogeneity, has led, in some cases, to even question the utility of social costs estimation [12,13]. Also recently, there have been critiques to the perspective adopted by social cost studies, arguing that a ‘public expenditure’ perspective should be taken instead [14,15]. Public expenditure studies focus solely on expenses incurred by governments. While this is of great importance from a government perspective and also for intermediate economic evaluations, public expenditure studies do not capture the whole cost, as is the case with social cost studies.

Therefore, despite methodological shortcomings and critiques, social cost estimation is an unavoidable task where informed decisions must be taken and limited resources allocated in an efficient manner. In this scenario, a clear need for methodological guidance protrudes as a first-order priority in the field, starting with the need to systematically assess the social costs of illegal drugs, alcohol and tobacco in the European Union. The aim of the present article is therefore to systematically review previous studies about social costs conducted in the European Union, focusing not only on cost estimates but also on the methodologies used.

Methods

Search strategy and inclusion criteria

This systematic review was made following the principles of the PRISMA statement [16]. A protocol was designed by the first author and then shared with the rest of the authors who made the pertinent amendments. The electronic databases PubMed and Scopus (which contains EMBASE) were searched from 1990 onwards, with the search strategy displayed in Appendix A of Supporting Information. Search terms were retrieved from previous reviews and authors’ own suggestions. Briefly, terms referring to illegal drugs (such as drugs, illicit drugs, illegal drugs, marihuana, opioids, cocaine, heroin, amphetamine, methamphetamine, street drugs), alcohol (such as drinking, drinker, alcohol dependence, alcohol abuse, risky or hazardous drinking) and tobacco (such as smoker, smoking, tobacco, cigarette) were combined with terms referring to social costs (such as social costs, economic costs, cost of illness, burden of disease, economic evaluation, traffic crashes, property damage, productivity losses or premature mortality). The European Monitoring Centre for Drugs and Drug Addiction database and the Cochrane Library were also searched, combining similar terms. Grey literature search consisted of previous systematic reviews, bibliographies, expert communications and the consulting of experts in the field. Only studies in English were considered. Studies were first screened based on title and abstract by two independent reviewers (PB and JR). The full text of potentially eligible studies was retrieved to check against inclusion and exclusion criteria, which are outlined in Table 1. Disagreements between reviewers were resolved by consultation with an expert on the field. Searches were conducted between March and April 2015.

Data extraction

Standardised extraction forms were used to extract the following information for each study:

- 1) Methodological characteristics.
- 2) Cost components included in the analysis and its magnitude.
- 3) Total estimated cost.

Methodological characteristics

Methodological characteristics extracted from the identified literature were related to:

- 1) The approaches used in the cost estimates, that is, prevalence or incidence approach. Prevalence-based estimates generally measure the costs of substance use in the present and the past in a given

Table 1. Inclusion and exclusion criteria

Inclusion criteria	Exclusion criteria
• Studies that consider the social costs of drugs, alcohol or tobacco in EU countries • English language or English summary available • Study conducted beyond 1990	• Non-English language summary available • Conference abstract • No costs quoted in the result section • Conducted in specific population sub-groups, such as pregnant women or adolescents • Not an original research article (i.e. review articles, systematic review articles and editorials) • Further publications of single studies

year, while incidence-based studies generally estimate the present and future costs of substance use in a given year.

- 2) The methods used for estimating the cost of premature mortality (i.e. human capital or demographic approach). In the human capital approach, the lost value of a deceased worker's production is estimated by present earnings plus a discounted rate of future earnings. The demographic approach compares the actual population size and structure to that of an 'otherwise healthy' population, that is, an alternative population in which there were no drug-related deaths.
- 3) The inclusion of the potential positive health effects of alcohol drinking (i.e. using gross cost or net cost; not applicable to either drugs or tobacco studies).
- 4) The discount rate used for adjusting future monetary values. The use of a discount rate acknowledges the fact that any amount of money received any given year is worth more than the same amount received next year (even if there is no inflation) because this year's resources become available for investment purposes a year earlier and so produce interest receipts or profits a year earlier.

Cost components included in the analysis

Despite a high variability, we tried to broadly include all possible costs reported in the literature. A common initial approach is to divide costs into tangible and intangible costs. Tangible costs can be further divided into direct costs (costs measuring direct consumption of societal resources) and indirect costs (costs arising from lost or impaired productivity, where no actual payment is made).

Intangible costs try to value life beyond its contribution to material production. Hence, they represent pain, suffering and the deterioration of quality of life. This type of cost, when reduced or eliminated, does not yield resources that can be made available for other uses, and is less likely to be included in cost estimations.

In an effort to display all cost components, we followed the international guidelines of Single *et al.* [17] where costs are finally classified according to the following major types of costs:

- 1) Consequences to health and welfare systems, which includes treatment for substance abuse, treatment of comorbidities and prevention, research and welfare services.
- 2) Productivity costs (i.e. consequences to the workplace), which includes premature mortality and lost employment or productivity.
- 3) Law enforcement and criminal justice costs.
- 4) Other costs (e.g. property destruction, fire loss, fire prevention).
- 5) Intangible costs.

Total estimated cost of illegal drugs, alcohol and tobacco

For each included study, the total estimated cost in monetary value was presented as originally reported by the study.

In order to facilitate comparisons across studies (which means comparing across different countries and different years), we calculated the percentage of GDP and the price per capita that the total costs represented in 2014. To achieve this, costs from Eurozone countries that were published before the introduction of the Euro were inflated to 2002 using Eurostat and inflation.eu, converted into Euros using the average exchange rate in 2002 on Oanda and then inflated to 2014. Costs published in Euros were simply inflated to 2014 values. Costs in British pounds were inflated to 2014 and converted into Euros. This operation resulted in the final price in millions of Euros for 2014. Dividing this number by the 2014 GDP values found in Eurostat, we obtained the percentage of GDP.

To obtain price per capita, we applied purchasing power parity (PPP) to the 2014 total cost using the 2013 PPP rates from Eurostat. This transformation allows minimising biases introduced by international

comparisons of market exchanges. Then, we divided this amount by the total 2014 population in each country, resulting in the final price per capita for the year 2014.

Quality assessment

Despite the existence of a number of guidelines for critical appraisal of economic evaluations [18–20], none of them is specifically designed for social cost studies. Therefore, we opted for developing our own evaluation tool. As part of the same project, and in parallel to this study, a review of existing guidance documents for estimating social costs was conducted (see Agència de Qualitat i Avaluació Sanitàries de Catalunya et al., forthcoming at <http://www.alicerap.eu/social-costs-of-addiction/171-social-costs-of-addiction.html>). The review concluded that the following items had to be evaluated in order to assess the quality of social cost estimation studies: theoretical framework (cost of illness versus utility evaluation methods); inclusion or not of private costs; methods for the feasible minimum calculation arcadian normal or exposure based comparators versus epidemiologic-distributional approach with scenario analysis); methods for costs estimation (human capital and prevalence approach versus willingness to pay and incidence approach); inclusion or not of intangibles; and cost categories included.

Instead of a numeric value, two categories were established according to the assessment of the different items. The first could be considered as a minimum standard which researchers should refer to for conducting

social cost estimation studies. The second framework would represent the ideal approach, allowing for a more comprehensive and high-quality estimation. Their detailed structure and components can be seen in Appendix B of the Supporting Information. Therefore, for each included study in our review, one of the authors assessed whether they conformed to the optimal, minimum or neither of the categories. In case of doubt the rest of the authors were consulted. No sensitivity analyses according to quality were conducted.

Results

Literature search

The electronic search strategy resulted in approximately 18 000 hits. After assessing title and abstract, 86 studies were selected for full review. The assessment of previous systematic reviews and other data sources yielded 28 more studies for full review. After exclusion criteria were applied, a total of 45 studies, reported in 43 publications, were finally included in this review (11 for drugs, 26 for alcohol and 8 for tobacco). Flow diagram is shown in Figure 1.

Methodological characteristics

Tables 2 to 5 outline the general methodological characteristics of the included studies. Most of the studies included in the three categories were prevalence-based studies (9 out of 11 studies for drugs, 21 out of 26 for

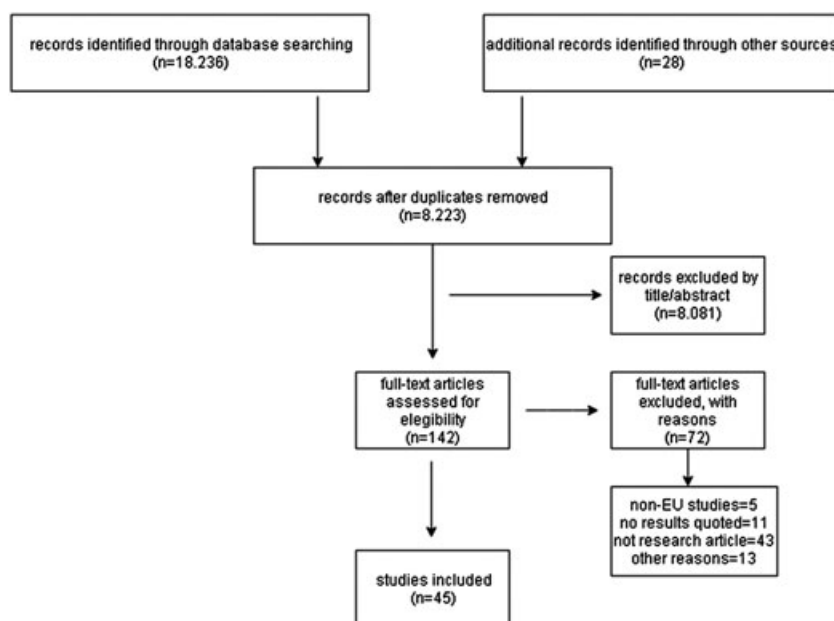


Figure 1. Study flow diagram.

Table 2. Methodologies of included studies

Country and year	Cost estimates	Premature deaths	Intangibles	Discount rate	Gross/net
Illegal drugs studies					
UK 1995 [21]	Bottom up	NA	Not included	NA	NA
France 1997 [22]	Prevalence	HCA	Not included	6%	NA
Spain 1997 [23]	Prevalence	HCA	Not included	NA	NA
Luxembourg 1999 [24]	Prevalence	NA	Not included	NA	NA
Sweden 2002 [25]	Prevalence	NA	Not included	NA	NA
Netherlands 2003 [26]	Prevalence	NA	Not included	NA	NA
Germany 2006 [27]	Prevalence	NA	Not included	NA	NA
Latvia 2008 [28]	Prevalence	NA	Not included	NA	NA
Belgium 2008 [29]	Prevalence	NA	Not included	NA	NA
Portugal 2010 [30]	NA	NA	NA	NA	NA
Finland 2012 [31]	Prevalence	NA	Not included	NA	NA
Alcohol studies					
Finland 1990 [32]	Prevalence	NA	Included	NA	Gross
Slovak Republic 1994 [33]	NA	NA	Not included	NA	Gross
Germany 1995 [34]	NA	NA	Not included	NA	Gross
Portugal 1995 [35]	Prevalence	HCA	Not included	5%	Gross
France 1996 [36]	Prevalence	NA	Not included	NA	NA
France 1997 [22]	Prevalence	HCA	Not included	6%	Gross
Spain 1998 [37]	NA	NA	Not included	NA	Gross
Sweden 1998 [38]	Prevalence	DA	Not included	NA	Gross
Switzerland 1998 [39]	Prevalence	HCA	Included (WTP)	0–2–6%	Net
Belgium 1999 [40]	NA	NA	Not included	NA	Net
Ireland 1999 [41]	NA	NA	Not included	NA	Gross
England & Wales 2001 [42]	Prevalence	HCA	Not included	NA	Gross
Norway 2001 [43]	Prevalence	HCA	Not included	3, 5%	Gross
Scotland 2001–2 [44]	Prevalence	HCA and WTP	Not included	6%	Gross
Germany 2002 [45]	Prevalence	HCA	Not included	5%	Net
Slovenia 2002 [19]	Prevalence	HCA	Not included	NA	Gross
Sweden 2002 [46]	Prevalence	HCA	Included	3%	Net
Austria 2004 [47]	Prevalence	HCA	Not included	NA	Gross
Italy 2004 [48]	Prevalence	HCA	Not included	NA	Gross
Portugal 2005 [49]	Prevalence	NA	Not included	NA	Gross
UK 2005 [50]	Prevalence	NA	Not included	NA	Gross
Estonia 2006 [51]	Prevalence	HCA	Not included	2–10%	Net
España 2007 [52]	Prevalence	HCA	Not included	NA	Gross
Ireland 2007 [53]	Prevalence	WTP	Not included	NA	Gross
Scotland 2007 [54]	Prevalence	HCA	Not included	NA	Gross
Scotland 2009–10 [55]	Prevalence	HCA	Included, WTP	NA	Gross
Tobacco studies					
Germany 1993 [56]	Prevalence	HCA	Not included	3%	NA
Denmark 1995 [57]	Incidence	HCA	Not included	NA	NA
Germany 1996 [58]	Prevalence	HCA	Not included	4%	NA
France 1997 [22]	Prevalence	HCA	Not included	6%	NA
Sweden 2001 [59]	Prevalence	HCA	Not included	5%	NA
Germany 2003 [60]	Prevalence	HCA	Not included	3%	NA
UK 2005 [61]	Prevalence	NA	Not included	NA	NA
Germany 2008 [62]	Bottom-up	HCA	Not included	NA	NA

DA, demographic approach; HCA, human capital approach; NA, not applicable or not indicated; WTP, willingness to pay approach.

alcohol, 6 out of 8 for tobacco). The remaining either used a bottom-up approach (which instead of a societal perspective follows a cohort of users over time and measures the costs they incur) or did not report the methodology employed.

For premature death estimates, there was a clear predominance of the human capital approach. In the

illegal drug studies, 2 followed this approach (9 did not report any method). For alcohol, 15 studies were based on it (regarding the rest of studies, 9 did not report on any method, 1 used both the demographic and willingness to pay approaches, and 1 used willingness to pay approach). For tobacco, 7 out of 8 used the human capital approach, with 1 study not reporting

Table 3. Comparison of drug cost estimates, all referring to 2014

Country and year	Final price millions of Euros	Final price with PPP applied	Price per capita in Euros	% of GDP
UK 1995	21.55	23.3	0.37	9.70E – 0.4%
France 1997	2666.49	2357.5	36.5	0.12%
Spain 1997	1047.51–1582.90	1163.1–1757.6	24.7–37.3	0.1%–0.15%
Luxembourg 1999	30.1	24.9	46.5	0.19%
Sweden 2002	592.53–1659.09	50.8–142.2	5.3–14.8	0.14%–0.39%
Netherlands 2003	2611.11	2379.2	141.6	0.40%
Germany 2006	5824.63–6821.12	5540.7–6488.6	67–78.5	0.2%–0.23%
Latvia 2008	118.4	174.3	89.9	2.75%
Belgium 2008	326.27	289.8	26	0.08%
Portugal 2010			24.5	
Finland 2012	383.83–457.64	310.3–370	56.4–67.3	0.58%–0.7%

GDP, gross domestic product; PPP, purchasing power parity.

Table 4. Comparison of alcohol cost estimates, all referring to 2014

Country and year	Final price in millions of Euros	Final price with PPP applied	Price per capita in Euros	% of GDP
Finland 1990	2141.2	1735.7	318.9	1.05%
Slovak Republic 1994	1030.3	1516.3	278	1.37%
Germany 1995	26 796	25 489.9	308.4	0.92%
Portugal 1995	658.7	844.7	79.6	0.38%
France 1996	2347.6–3368.1	2075.6–2977.8	32.1–46.1	0.15%
France 1997	23 060.7	20 388.7	315.4	1.07%
Spain 1998	5668.8	6294.3	133.7	0.54%
Sweden 1998	14.9	1.3	1548.4	3.47%
Switzerland 1998	1628.1	891.3	109.3	0.32%
Belgium 1999	5991.4	5321.4	477.5	1.49%
Ireland 1999	2699.1	2449	523.6	1.46%
England and Wales 2001	30 086.4–32 567.3	32 522.1–35 203.9	512.3–554.5	1.46–1.47%
Norway 2001	2123.8–2390.8	174.2–196.1	34.2–38.5	0.56–0.63%
Scotland 2001–2002	1710.6	1849.1	347.1	1.06%
Germany 2002	29 556.4	28 115.7	340.2	1.02%
Slovenia 2002	0.104	0.13	0.062	0.28%
Sweden 2002	2721.1	233.3	24.2	0.63%
Austria 2004 (alcohol + drugs)	1767.3	1581.4	185.5	0.54%
Italy 2004 (alcohol + drugs)	1750	1735	28	0.11%
Portugal 2005	211	271	26	0.12%
UK 2005	4549	4917	77	0.20%
Estonia 2006	275.1–387.3	377.3–531.2	293.9–413.8	1.41–2.72%
Spain 2007	3038.6	3373.9	71.7	0.29%
Ireland 2007	3781.7	3431.4	733.6	2.04%
Scotland 2007	3259.6	3523.5	661.4	2.03%
Scotland 2009–2010	1811.2	1957.8	367.5	1.13%

GDP, gross domestic product; PPP, purchasing power parity.

on any methods regarding premature mortality estimates.

Most alcohol studies used gross costs (i.e. did not take into account the potential beneficial effects of consumption). Out of 26, only 5 used a net cost perspective, whereas 1 study did not report on this item.

The discount rate applied in the drugs studies turned out to be 6%, belonging to a single study, with the other 10 studies not reporting on this item. Alcohol studies ranged from 0% to 10% (the number of studies not reporting on this item was 18), and tobacco studies ranged from 3% to 6%, with 3 studies not reporting on this item.

Table 5. Comparison of tobacco cost estimates, all referring to 2014

Country and year	Final price in millions of Euros	Final price with PPP applied	Price per capita in Euros	% of GDP
Germany 1993	23 848.3	22 685.8	274.5	0.82%
Denmark 1995	777.6	76.5	13.6	0.30%
Germany 1996	21 691.6	20 634.3	249.7	0.75%
France 1997	17 827.6	15 761.9	243.8	0.83%
Sweden 2001	1185.5	101.6	10.6	0.28%
Germany 2003	25 210.6	23 981.8	290.2	0.87%
UK 2005	10 717.7	11 585.4	182.5	0.48%
Germany 2008	33 987.1	32 330.5	391.2	1.17%

GDP, gross domestic product; PPP, purchasing power parity.

Cost components included in the studies

Tables S1 to S3 outline cost components included in the different studies, according to the classification suggested by Single [17]. While there is a clear and constant inclusion of direct costs related to treatment of substance abuse and its comorbidities for all three categories, it is difficult to find a consistent pattern for the rest of the items.

Most of the studies in the drugs section include costs related to law enforcement and criminal justice (9 studies), with research and prevention costs being also frequently included (8 studies). Regarding costs because of premature mortality or lost productivity, they are only taken into account in 3 studies.

Alcohol studies tend to be the most inclusive. Besides healthcare costs, included in all of them, 6 studies evaluate research and prevention costs, 9 studies also focus on welfare and social services related expenses, while 12 studies also include other costs, such as property damage or accidents. A great majority of studies include indirect costs, with 19 studies referring to premature mortality and 19 studies as well focusing on lost employment or productivity. Fourteen studies assessed criminal justice and law enforcement costs.

All tobacco studies assessed costs related to healthcare, while only one included expenses in research and prevention. Indirect costs were measured in the majority of them; premature mortality was included in 4 studies and lost productivity in 5 studies.

Finally, intangible costs were rarely included in social cost studies, with only 3 in the alcohol domain including them.

Cost estimates

Cost estimates as reported by studies, including the percentage of GDP (if reported), the share between direct and indirect costs, and the total cost estimate for the study year can be seen in Tables S4 to S6. For each section there are two tables. Tables 2 to 5 enable study comparison, outlining total cost in 2014, percentage of GDP

in 2014 and price per capita in 2014, following the operations as explained in section 2.2.3.

In the illegal drug section, a wide range of results is observed. For example, price per capita in 2014 ranged from €0.37 in the UK to €78 in Germany. The same happened with percentage of GDP, which ranged from 0.001% in the United Kingdom to 0.4% in the Netherlands. Although only 2 studies included indirect costs, results between them were also quite heterogeneous, with one estimating their share in the total costs as 24.4% (Spain) and the other as 45.7% (France). The share of direct costs ranged from 54.3% (France) to 100% (this number belonging to studies not taking into account any indirect costs).

Similarly, alcohol estimates presented a wide range of estimations, with price per capita ranging from €26 (Portugal) to €1500 (Sweden). The percentage of GDP ranged from 0.11% (Italy) to 3.47% (Sweden). The share of indirect costs ranged from 13.5% (Scotland) to 87.2% (Spain). For direct costs, the share ranged from 5.7% (Scotland) to 80% (Belgium). Although only included in 2 studies, intangible costs accounted for the majority of the costs (66% and 81%, for Switzerland and Scotland respectively). Usually, indirect costs were higher than direct costs for alcohol studies (in 14 of 26 studies).

Tobacco studies also showed a high degree of variability in results. Price per capita ranged from €10.55€ (Sweden) to €391 (Germany). Percentage of GDP ranged from 0.28% (Sweden) to 1.17% (Germany). The share of indirect costs ranged from 12.2% (Denmark) to 74% (Sweden). Direct costs ranged from 26% (Sweden) to 87.8% (Denmark). None of the studies estimated intangible costs. In half of the studies, indirect costs were larger than direct costs, while 1 remaining study did not assess any indirect costs (United Kingdom).

Quality assessment

Only 2 studies in the illegal drugs domain, 3 in the tobacco, and 8 in the alcohol were deemed as conforming

to the standards of the minimum desirable framework recommended by AQuAS *et al.* (forthcoming). No study was classified as conforming to the optimal framework. The classifications can be seen in Appendix C of the Supporting Information.

Discussion

Several methodological shortcomings were identified during this review, especially a high methodological heterogeneity and a global poor quality of included studies. Therefore, results should be interpreted with caution. That being said, the data analysed suggest that social costs of illegal drug, alcohol and tobacco use impose a heavy economic burden on the EU.

We covered the whole spectrum of substance use; hence, we included studies assessing illegal drugs, alcohol and tobacco-related costs. However, some important differences must be pointed out.

The first is the almost universal illegal status of drugs, which turns drugs into a different kind of market, where data availability could potentially be more difficult to obtain or to correctly organise [63,64]. This could be one of the reasons why alcohol and tobacco studies tend to be more inclusive than drug studies, in respect to parameters and types of costs included. Globally, lack of primary data is one of the biggest weaknesses of social costs' estimations. When this cannot be directly addressed, it is suggested that authors use other sources or proxies, such as existing data from similar countries. Although this method adds an important degree of imprecision, it would prevent the omission of essential cost components, such as mortality or lost productivity [11,17].

Another important observation is how costs are distributed between direct costs (i.e. costs that arise because of the direct destruction or consumption of resources, such as medical expenses or property damage) and indirect costs (i.e. costs that arise because of non-production of future resources). While there is a clear predominance of indirect costs in terms of total share for alcohol studies, and in the tobacco section there seems to be a tendency to equality, in the drugs studies it is difficult to state any conclusion, because only 2 out of 8 studies evaluated indirect costs.

Besides different theoretical frameworks (cost of illness vs bottom-up approaches), other relevant aspects of the studies' methodologies are highly heterogeneous. For example, most of the studies are based on a prevalence approach, although some employ an incidence-based approach. Regarding premature death cost estimates, most of the studies use the human capital approach, but some rely on the 'willingness-to-pay' approach. Another huge difference is observed in the discount rates applied, which vary greatly. The assumptions these different

approaches are based on do not necessarily have to be contradictory, but rather complementary. What becomes worrisome is that the validity of comparisons between countries becomes compromised. Do the differences in costs reflect real differences or are they just a by-product of the different methodologies applied? This difficulty has in fact been found in previous comparisons of social cost estimates between countries [65,66], and will remain unanswered until uniform guidelines for the estimation of social costs are adopted in the EU.

The methodological difficulties in social cost estimation studies have led some to argue in favour of public expenditure studies instead, suggesting it would facilitate a common methodological framework [14,15]. However, as previously stated, public expenditure studies do not aim at capturing the whole cost, as it is the case with social cost studies. Indeed, it has been shown that public, that is, labelled drug-related, expenditure is highly correlated with nations' wealth [67]. Therefore, public expenditure studies might be highly misleading when trying to assess the global impact that drug use has on society as a whole.

Focusing on major gaps regarding economic research, it becomes evident that the alcohol field is ahead of that of tobacco and drugs. Not only are there a larger number of alcohol studies, but also they are more inclusive and thorough in their investigations, with more components and parameters included in their estimates. It should, therefore, constitute a reference point for countries and policy makers undertaking social cost estimations of tobacco, and especially illegal drugs, where the state of affairs might be more behind.

The time frame covered by this review is between 1990 and 2014. It means that some studies can be considered recent, while others might date back as far as 25 years. Treatment practices change over time, as do the illicit drug markets, with new drugs becoming available. Also, new treatments become available for substance use. Taken together, all these suggest that caution should be the norm when extrapolating cost results from the past to the present.

Despite all these difficulties, an attempt to conduct a comparison between studies was done by using economic indices, such as inflation and PPP. Despite the observed heterogeneity, both in methodologies and in the range of results, costs can be considered high. It means that an important share of national economies is consumed by substance use, either directly or indirectly. For drug studies, most of their estimations ranged between 0.1% and 0.4% of GDP. Considering the scarcity of data availability in this field, it would not be unwise to consider real costs even higher. For tobacco, studies ranged from 0.3% to 1.2% of GDP, and for alcohol figures were even higher despite a huge variability, with the majority of studies exceeding the 1% of GDP and some of them between 2% and 3.5% of GDP. Taken together, all these numbers

clearly indicate the deleterious effect that substance use has on society at large.

Limitations

Several limitations must be taken into account when interpreting the findings of this review. Whatever their theoretical framework, social costs studies heavily rely on many assumptions that might not be accurate enough. Therefore, they must be considered a rough approximation rather than an exact estimation of reality. In this sense, their quality is highly dependent on the capacity for data extraction in many areas (e.g. criminal justice system, insurance companies, healthcare system). This capacity might be very different across countries, with some of them having a relative tradition of data storage, management and extraction, whereas others might be rather new in this task.

Another relevant caveat is the existence of a broad methodological heterogeneity. This means that for a given reality, different estimations can be obtained. The problematic consequence of this fact is that it makes direct comparisons between countries difficult, and jeopardises the investigation of one of the aims of social cost estimation, that is, the assessment of cost-effectiveness policies. Also, methodological recommendations change over time. For example, in the last guidelines of the World Health Organization for cost estimations in the alcohol field [68], it is suggested that the human capital approach should not be used, and instead, only friction costs (costs derived of the need to replace a worker) should be included. None of the studies included in this review, however, estimated indirect costs with the friction costs method.

The intrinsically complex causal relationships between substance use and its deleterious effects and costs must also be taken in account. For example, when estimating the premature mortality of alcohol patients, one should take into account other relevant factors, such as smoking, and the relevant interactions between alcohol and tobacco. Illegal drugs, in their turn, suffer specially from such interactions, because comorbidities with alcohol and tobacco are extremely frequent. Should we consider one as causing the other, or should we consider them as comorbidities without causal hierarchy? Although some methods exist for incorporating this complexity into the studies, such as attributional fractions, they are far from perfect.

Data availability is a huge limitation for social cost estimates. Nations do not usually have all the information required for such studies, and hence have to rely on different approximations to close the data gaps. Although the illegal status of drugs might not suppose a larger data gap in comparison to tobacco and alcohol, there is no

doubt that illegal drugs constitute a huge market on their own, but at present, researchers are hardly able to evaluate it [64].

Regarding the limitations of this review, we tried to be as systematic as possible and opted for a highly sensitive search in order to not miss relevant studies. This strategy meant a huge number of references had to be reviewed. While it could be considered as a limitation in the specificity of the search strategy, we prioritised sensibility. That being said, there remains the possibility of having missed relevant studies, especially those not available in English. Our main focus of study was the EU zone. While this might have granted the review a higher internal validity, it imposes several limitations when trying to extrapolate our results out of the EU. Also relevant, the quality assessment of this review was based on the development of a new tool, which has not been previously validated.

Conclusions

Given all the limitations identified throughout this review, firm conclusions are difficult to be drawn. While the exact economic impact of substance use cannot at this stage be accurately determined, the reviewed studies seem to indicate that there exists an impact of substance use on society from an economic perspective. Illegal drugs, alcohol and tobacco keep consuming nations' resources in many ways, such as increasing costs of healthcare delivery, law and crime enforcement or lost productivity. It seems that they could, all together, add up to become a significant share of nations' GDP. Hence, from a public health and policy maker perspective, it seems reasonable to state that there is a strong need to address these issues. Given our results, it seems that in order to unambiguously progress in this task, there is an urgent need for methodological guidance, a need to standardise and homogenise the methodologies employed for social costs evaluations.

Acknowledgments

The research leading to these results or outcomes has received funding from the European Union's Horizon 2020 Programme (H2020/2014-2020), under Grant Agreement no. JUST/2013/DPIP/AG 4755—Looking at Economic Analyses of Drugs and Economic Recession (LEADER—www.leader-project.net). Antoni Gual has received research funding from Lundbeck, D&A Pharma and TEVA, not related to the present study, and honoraria from Lundbeck, D&A Pharma and AbbVie, not related to the present study. Pablo Barrio has received honoraria from Lundbeck, not related to the present study.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Table S1. Cost components included: drugs studies.

Table S2. Cost components included: alcohol studies.

Table S3. Cost components included, tobacco studies.

Table S4. Drugs cost estimates reported by studies.

Table S5. Alcohol cost estimates reported by studies.

Table S6. Tobacco cost estimates reported by studies.

Appendix A. Search strategy.

Appendix B. Proposed estimation frameworks.